

MA 587: Homework 2

Joseph F. Specht IV

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1 Linear Interpolation on the Uniform Grid

1.A Bounding Max Error of u'

I used Gemini in this section to remind me of the Taylor theorem with a remainder.

1. Begin by Taylor expanding $u(x_j)$ and $u(x_{j-1})$ about x with the remainder on the second derivative.

We only know u to be twice continuously differentiable on the interval, so we attach the reminder term (ξ_i) to the second derivative.

$$u(x_j) = u(x) + u'(x)(x_j - x) + \frac{1}{2}u''(\xi_1)(x_j - x)^2 \quad (1)$$

$$u(x_{j-1}) = u(x) + u'(x)(x_{j-1} - x) + \frac{1}{2}u''(\xi_2)(x_{j-1} - x)^2 \quad (2)$$

2. Subtract the two expressions.

$$u(x_j) - u(x_{j-1}) = [u(x) - u(x)] + u'(x)[(x_j - x) - (x_{j-1} - x)] + \frac{1}{2}[u''(\xi_1)(x_j - x)^2 - u''(\xi_2)(x_{j-1} - x)^2] \quad (3)$$

3. Recognize $u(x_j) - u(x_{j-1}) = hu'_h(x)$, $x_j - x_{j-1} = h$, and divide through by h .

$$\tilde{u}'_h(x) = \frac{1}{h}u'(x)h + \frac{1}{2h}[u''(\xi_1)(x_j - x)^2 - u''(\xi_2)(x_{j-1} - x)^2] \quad (4)$$

$$\tilde{u}'_h(x) = u'(x) + \frac{1}{2h}[u''(\xi_1)(x_j - x)^2 - u''(\xi_2)(x_{j-1} - x)^2] \quad (5)$$

$$\tilde{u}'_h(x) - u'(x) = \frac{1}{2h}[u''(\xi_1)(x_j - x)^2 - u''(\xi_2)(x_{j-1} - x)^2] \quad (6)$$

4. Take the absolute value of both sides.

$$|\tilde{u}'_h(x) - u'(x)| = \left| \frac{1}{2h} [u''(\xi_1)(x_j - x)^2 - u''(\xi_2)(x_{j-1} - x)^2] \right| \quad (7)$$

5. Apply the triangle inequality.

$$|\tilde{u}'_h(x) - u'(x)| \leq \frac{1}{2h} [|u''(\xi_1)| (x_j - x)^2 + |u''(\xi_2)| (x_{j-1} - x)^2] \quad (8)$$

6. Let's say the point $y \in [0, 1]$ is where $u''(x)$ has the highest absolute value in the entire domain. We know the maximum change the derivative of our function can undergo is when the second derivative is at its maximum absolute value. Therefore, we assume we are at y and factorize the expression.

$$|\tilde{u}'_h(x) - u'(x)| \leq \frac{1}{2h} \max_{y \in [0, 1]} |u''(y)| [(x_j - x)^2 + (x_{j-1} - x)^2] \quad (9)$$

7. We also know x is somewhere between x_j and x_{j-1} , so $x_j - x \leq x_j - x_{j-1} = h$ and $x - x_{j-1} \leq x_j - x_{j-1} = h$. We can substitute these relations to find an upper bound.

$$(x_j - x)^2 + (x_{j-1} - x)^2 \leq h^2 + h^2 = 2h^2 \quad (10)$$

$$|\tilde{u}'_h(x) - u'(x)| \leq \frac{2h^2}{2h} \max_{y \in [0, 1]} |u''(y)| \quad (11)$$

8. Arriving at the final expression.

$$|u'(x) - \tilde{u}'_h(x)| \leq h \max_{y \in [0, 1]} |u''(y)| \quad (12)$$

1.B Boundary Value Problem

1. Begin by using the hint with the given boundary conditions. Note, u_h inherits the boundary conditions from u as $u_h \in V_h \subset V$ (I don't know if it should be $\subset$ or $\subseteq$).

$$\int_0^x (u(x) - u_h(x))'(s) ds = (u(x) - u_h(x)) - (u(0) - u_h(0)) = (u(x) - u_h(x)) - (0 - 0) \quad (13)$$

$$\int_0^x (u(x) - u_h(x))'(s) ds = u(x) - u_h(x) \quad (14)$$

2. Take the absolute value of both sides.

$$|u(x) - u_h(x)| = \left| \int_0^x (u(x) - u_h(x))'(s) ds \right| \quad (15)$$

3. Apply Cauchy-Schwartz assuming one of the vectors under the integral is 1.

$$|u(x) - u_h(x)| \leq \sqrt{\int_0^x 1^2 ds} \sqrt{\int_0^x |(u(x) - u_h(x))'(s)|^2 ds} \quad (16)$$

$$|u(x) - u_h(x)| \leq \sqrt{x} \sqrt{\int_0^x |(u(x) - u_h(x))'(s)|^2 ds} \quad (17)$$

4. We know $u \in [0, 1]$, so we know $\sqrt{x} \leq 1$.

$$|u(x) - u_h(x)| \leq \sqrt{\int_0^x |(u(x) - u_h(x))'(s)|^2 ds} \quad (18)$$

5. Recognize the right hand side as $|u - u_h|$.

$$|u(x) - u_h(x)| \leq |u(x) - u_h(x)| \quad (19)$$

6. Use the relation we proved in part a.

$$|u(x) - u_h(x)| \leq |u(x) - u_h(x)| \leq h \max_{y \in [0,1]} |u''(y)| \quad (20)$$

$$|u(x) - u_h(x)| \leq h \max_{y \in [0,1]} |u''(y)| \quad (21)$$

2 Hilbert Space

2.A Equivalence of (M) and (V)

1. Let's say $u \in V$ is the solution to our problem. If we have some vector $v \in V$, we can express $v = u + w$ as the sum of the optimal solution u and a deviation from the optimal solution w .

$$F(v) = F(u + w) = \frac{1}{2}a(u + w, u + w) - L(u + w) \quad (22)$$

2. Expand both terms on the RHS keeping in mind $a(\cdot, \cdot)$ is bilinear and $L(\cdot)$ is linear.

$$a(u + w, u + w) = a(u, u) + 2a(u, w) + a(w, w) \quad (23)$$

$$L(u + w) = L(u) + L(w) \quad (24)$$

3. Substitute.

$$F(v) = \frac{1}{2}[a(u, u) + 2a(u, w) + a(w, w)] - [L(u) + L(w)] \quad (25)$$

4. Realize the follow relationships. Use the variation form (V) for the second relationship.

$$F(u) = \frac{1}{2}a(u, u) - L(u) \quad (26)$$

$$a(u, w) = L(w) \Rightarrow a(u, w) - L(w) = 0 \quad (27)$$

5. Apply relationships.

$$F(v) = F(u) + \frac{1}{2}a(w, w) \quad (28)$$

6. Because a is coercive $\exists \alpha > 0$ such that $a(w, w) \geq \alpha \|w\|_v^2$.

$$\frac{\alpha}{2}\|w\|_v^2 + F(u) \leq \frac{1}{2}a(w, w) + F(u) = F(v) \quad (29)$$

7. We are given $F(u) \leq F(v)$. If $\|w\|_v^2 \neq 0$, $F(u) > F(v)$, which breaks the given inequality.

$$\therefore \|w\|_v^2 = 0 \quad (30)$$

$$F(u) \leq F(v) \quad (31)$$

2.B Energy Norm

1. Start by expanding the RHS of the given inequality. Additionally, "add zero" by adding and subtracting u_h at the same time. Remembering the definition for v , also define $A \equiv u - u_h$, $B \equiv u_h - v$.

$$\|u - v\|_a^2 = a(u - v, u - v) = a([u - u_h] + [u_h - v], [u - u_h] + [u_h - v]) \quad (32)$$

$$= a(A, A) + 2a(A, B) + a(B, B) \quad (33)$$

2. The variational form states $a(u, w) = L(w) \forall w \in V$. Similarly, $a(u_h, w) = L(w) \forall w \in L_h$. As, $V_h \subset V$, V_h inherits the properties of V .

$$a(u, w) - a(u_h, w) = L(w) - L(w) \quad (34)$$

$$a(u, w) - a(u_h, w) = 0 \quad \forall w \in V_h \quad (35)$$

3. Using the above relation, we see $2a(u - u_h, u - v) = 0$ as $u - v = w$.

$$\|u - v\|_a^2 = a(u - u_h, u - u_h) + 0 + a(u_h - v, u_h - v) \quad (36)$$

4. Recognize the following.

$$\sqrt{a(v, v)} = \|v\|_a \quad (37)$$

$$a(v, v) = \|v\|_a^2 \quad (38)$$

5. Rewrite the expression using the above relations.

$$\|u - v\|_a^2 = \|u_h - v\|_a^2 + \|u - u_h\|_a^2 \quad (39)$$

6. We know a is coercive such that $a(u_h - v, u_h - v) \geq \alpha \|u_h - v\|_v^2$, which is always ≥ 0 , but ≤ 1 from the previous relationship.

$$\therefore \|u - u_h\|_a \leq \|u - v\|_a \quad (40)$$

3 2D FEM Element Basis

3.A Node Conditions

We know the three points cannot be colinear otherwise there is not a unique plane that can describe the three points. For a mathematical description, we can use the notation from part b to state the following:

$$\det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \neq 0 \quad (41)$$

If the determinant was equal to zero, the matrix would be singular and thus not span $\mathbb{R}^2$. In other words, the points could be defined by an infinite number of planes, which is not indicative of a triangle.

3.B Basis Functions

1. Begin with ϕ_1

$$\phi_1(N_1) = 0 + 0 + c = 1 \quad (42)$$

$$\phi_1(N_2) = ha + 0 + c = 0 \quad (43)$$

$$\phi_1(N_3) = 0 + hb + c = 0 \quad (44)$$

2. ϕ_2

$$\phi_2(N_1) = 0 + 0 + c = 0 \quad (45)$$

$$\phi_2(N_2) = ha + 0 + c = 1 \quad (46)$$

$$\phi_2(N_3) = 0 + hb + c = 0 \quad (47)$$

3. Finally, ϕ_3

$$\phi_3(N_1) = 0 + 0 + c = 0 \quad (48)$$

$$\phi_3(N_2) = ha + 0 + c = 0 \quad (49)$$

$$\phi_3(N_3) = 0 + hb + c = 1 \quad (50)$$

4. These systems of equations can be solved to find the basis vectors as such.

$$\phi_1(x, y) = -\frac{x}{h} - \frac{y}{h} + 1 \quad (51)$$

$$\phi_2(x, y) = \frac{x}{h} \quad (52)$$

$$\phi_3(x, y) = \frac{y}{h} \quad (53)$$

5. Any linear function can be written as $\phi(x, y) = ax + by + c$. The three basis vectors for this problem are $x, y, 1$. If we sum all ϕ_i , we see the following.

$$\phi_1 + \phi_2 + \phi_3 = 1 \quad (54)$$

6. We can plainly see we can write any $\phi(x, y) = ax + by + c$ as the following. This shows our basis vectors span the domain.

$$\phi(x, y) = ax + by + c = \alpha\phi_1 + \beta\phi_2 + \gamma[\phi_1 + \phi_2 + \phi_3] \quad (55)$$

3.C Stiffness Matrix

1. We need the following gradients.

$$\nabla\phi_1 = \begin{bmatrix} -1/h \\ -1/h \end{bmatrix} \quad (56)$$

$$\nabla\phi_2 = \begin{bmatrix} 1/h \\ 0 \end{bmatrix} \quad (57)$$

$$\nabla\phi_3 = \begin{bmatrix} 0 \\ 1/h \end{bmatrix} \quad (58)$$

2. We also need the following dot products. Note, the dot product is linear.

$$\nabla\phi_1 \cdot \nabla\phi_1 = 2/h^2 \quad (59)$$

$$\nabla\phi_1 \cdot \nabla\phi_2 = -1/h^2 \quad (60)$$

$$\nabla\phi_1 \cdot \nabla\phi_3 = -1/h^2 \quad (61)$$

$$\nabla\phi_2 \cdot \nabla\phi_3 = 0 \tag{62}$$

3. We see each entry for A is given as:

$$a_{ij} = \int_K \nabla u \cdot \nabla v dx \tag{63}$$

4. All of our dot products are constants, so move them out of the integral.

$$a_{ij} = \nabla u \cdot \nabla v \int_K dx \tag{64}$$

5. We are told $\int_K dx$ is the area of the triangle. We also know the area is given as $1/2$ the base times height.

$$\int_K dx = \frac{h^2}{2} \tag{65}$$

6. We notice all of the $1/h^2$ from the dot products cancel with the area. This leaves us to form our matrix for A as such:

$$\begin{bmatrix} 1 & -1/2 & -1/2 \\ -1/2 & 1/2 & 0 \\ -1/2 & 0 & 1/2 \end{bmatrix} \tag{66}$$

7. This is the expected form.

4 Sobolev Space

I used Gemini to refresh me on what a Sobolev Space is.

4.A Sequential Norms

1. We have the following.

$$\|u\|_{H^s} = \sum_{|\alpha| \leq s} \|D^\alpha u\|_{L^2}^2 \quad (67)$$

$$\|u\|_{H^k} = \sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^2}^2 = \sum_{|\alpha| \leq s} \|D^\alpha u\|_{L^2}^2 + \sum_{s \leq |\alpha| \leq k} \|D^\alpha u\|_{L^2}^2 \quad (68)$$

2. Because $\sum_{s \leq |\alpha| \leq k} \|D^\alpha u\|_{L^2}^2$ is a set of norms, we know each norm ≥ 0 . Therefore, we can write:

$$\sum_{|\alpha| \leq s} \|D^\alpha u\|_{L^2}^2 \leq \sum_{|\alpha| \leq s} \|D^\alpha u\|_{L^2}^2 + \sum_{s \leq |\alpha| \leq k} \|D^\alpha u\|_{L^2}^2 = \sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^2}^2 \quad (69)$$

$$\sum_{|\alpha| \leq s} \|D^\alpha u\|_{L^2}^2 \leq \sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^2}^2 \quad (70)$$

3. Express the sums as the inner-product induced norm.

$$\sum_{|\alpha| \leq s} \|D^\alpha u\|_{L^2}^2 = \|u\|_{H^s}^2 \quad (71)$$

$$\sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^2}^2 = \|u\|_{H^k}^2 \quad (72)$$

4. Finally,

$$\|u\|_{H^s} \leq \|u\|_{H^k} \quad (73)$$

4.B Triangle Inequality

1. Write the following.

$$\|u + v\|_{H^k}^2 = \sum_{|\alpha| \leq k} \|D^\alpha(u + v)\|_{L^2}^2 \quad (74)$$

2. Derivatives are linear operators.

$$\sum_{|\alpha| \leq k} \|D^\alpha(u + v)\|_{L^2}^2 = \sum_{|\alpha| \leq k} \|D^\alpha(u) + D^\alpha(v)\|_{L^2}^2 \quad (75)$$

3. Use the triangle inequality.

$$\sum_{|\alpha| \leq k} \|D^\alpha(u) + D^\alpha(v)\|_{L^2}^2 \leq \sum_{|\alpha| \leq k} \|D^\alpha(u)\|_{L^2}^2 + \sum_{|\alpha| \leq k} \|D^\alpha(v)\|_{L^2}^2 \quad (76)$$

4. Use the first hint.

$$\sum_{|\alpha| \leq k} \|D^\alpha(u) + D^\alpha(v)\|_{L^2}^2 \leq \sum_{|\alpha| \leq k} \|D^\alpha(u)\|_{L^2}^2 + \sum_{|\alpha| \leq k} \|D^\alpha(v)\|_{L^2}^2 + 2 \sum_{|\alpha| \leq k} \|D^\alpha(u)\|_{L^2}^2 \|D^\alpha(v)\|_{L^2}^2 \quad (77)$$

5. Use the third hint to express the following:

$$\|a\|_2 = \sqrt{\sum (\text{comps of } a)^2} = \sqrt{\sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^2}^2} \quad (78)$$

6. Then we can rewrite the previous expression using the definition for the inner product induced norm

$$\|u + v\|_{H^k}^2 \leq \|u\|_{H^k}^2 + \|v\|_{H^k}^2 + 2 \sum_{|\alpha| \leq k} \|D^\alpha(u)\|_{L^2}^2 \|D^\alpha(v)\|_{L^2}^2 \quad (79)$$

7. Using the expression from the third hint and the Cauchy-Schwartz inequality.

$$\sum_{|\alpha| \leq k} \|D^\alpha(u)\|_{L^2}^2 \|D^\alpha(v)\|_{L^2}^2 \leq \|u\|_{H^k} \|v\|_{H^k} \quad (80)$$

8. Substitute.

$$\|u + v\|_{H^k}^2 \leq \|u\|_{H^k}^2 + \|v\|_{H^k}^2 + 2 \sum_{|\alpha| \leq k} \|D^\alpha(u)\|_{L^2}^2 \|D^\alpha(v)\|_{L^2}^2 \leq \|u\|_{H^k}^2 + \|v\|_{H^k}^2 + 2\|u\|_{H^k} \|v\|_{H^k} \quad (81)$$

$$\|u + v\|_{H^k}^2 \leq \|u\|_{H^k}^2 + \|v\|_{H^k}^2 + 2\|u\|_{H^k} \|v\|_{H^k} \quad (82)$$

$$\|u + v\|_{H^k}^2 \leq (\|u\|_{H^k} + \|v\|_{H^k})^2 \quad (83)$$

$$\|u + v\|_{H^k} \leq \|u\|_{H^k} + \|v\|_{H^k} \quad (84)$$